

The Learnability of Model-Theoretic Interpretation Functions in Artificial Neural Networks

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Outline

- 1 Introduction & Motivation
- 2 Framework
- 3 Models & Evaluation
- 4 Results
- 5 Discussion & Conclusion

The Central Question

Can neural networks learn to *interpret* language
in a way that **generalizes compositionally**?

Systematicity

Anyone who understands “the cat chased the mouse”
can also understand “the mouse chased the cat.”

We systematically extend learned interpretation abilities
to **out-of-training-sample** combinations of familiar elements.

(Fodor and Pylyshyn, 1988)

The Classical Argument

Fodor & Pylyshyn's challenge:

- Systematicity requires compositional symbolic representations
 - Neural networks can exhibit it only by *covertly implementing* a symbol system
- ⇒ Neural nets = implementation vehicles, not explanations

If that is right, neural-network models of language understanding are explanatorily vacuous.

Our question: how much systematicity can a neural interpretation function learn without implementing a symbol system (from structured world knowledge)?

(Fodor and Pylyshyn, 1988)

A Neural Alternative: Frank et al. (2009)

Frank et al. (2009) showed that:

- Simple Recurrent Networks (SRNs) can learn to map sentences to distributed truth-conditional representations
- These representations support *some* systematic generalization
- **Without** implementing symbolic structures

Key insight: systematicity can arise from *structure in the world* encoded in analogical situation vectors, not only from symbolic compositionality in mental representations.

But: only SRNs, a limited test set (~ 80 sentences), and an older MATLAB/Prolog implementation.

(Frank et al., 2009)

Interpretation as Encoding

Core mapping

discrete sentence $\rightarrow$ neural encoder $\rightarrow$ meaning vector in $[0, 1]^{150}$

- Not language modeling, classification, or sequence generation
- **Encoding task:** map linguistic input to continuous truth-conditional representations
- Vector similarity reflects co-occurrence in possible worlds, not just text co-occurrence
- Targets come from **model-theoretic world structure**, not corpus-based distributional statistics

Our Contributions

- 1 **Entity vectors**: extend representations to include entity-level information (type e)
Training targets: 150-dim (truth-conditional only) or 300-dim (+ entities)
- 2 **Modern architectures**: LSTM, GRU, Transformer (AbsPE / RoPE) alongside SRN
Capacity-matched at $\sim 66k$ parameters
- 3 **Principled competing events**: Z3 validation replaces hardcoded rules
- 4 **Extended tests**: $\sim 80 \rightarrow \sim 350$ test sentences across 4 groups
- 5 **Two-dimensional difficulty**: results disaggregated by modifier complexity

140 trained models ($7 \text{ architectures} \times \pm \text{entity} \times 2 \text{ splits} \times 5 \text{ seeds}$)

Python / PyTorch implementation:

https://github.com/abrsvn/neural_model_theoretic_interpretation_ACL_2026

Preview of Key Findings

Finding 1: Systematicity difficulty has two dimensions

Test type (Word > Sentence > Complex > Basic)

× modifier complexity (canonical > +location > +manner > +both)

Finding 2: Gating in recurrent networks is critical

GRU/LSTM ≫ Transformers on the hardest test (Basic Event)

Ungated SRN does *not* show the same Basic-Event advantage

Finding 3: Entity information helps most where it matters

Largest benefit on Basic Event (novel entity combinations);

gated architectures benefit most

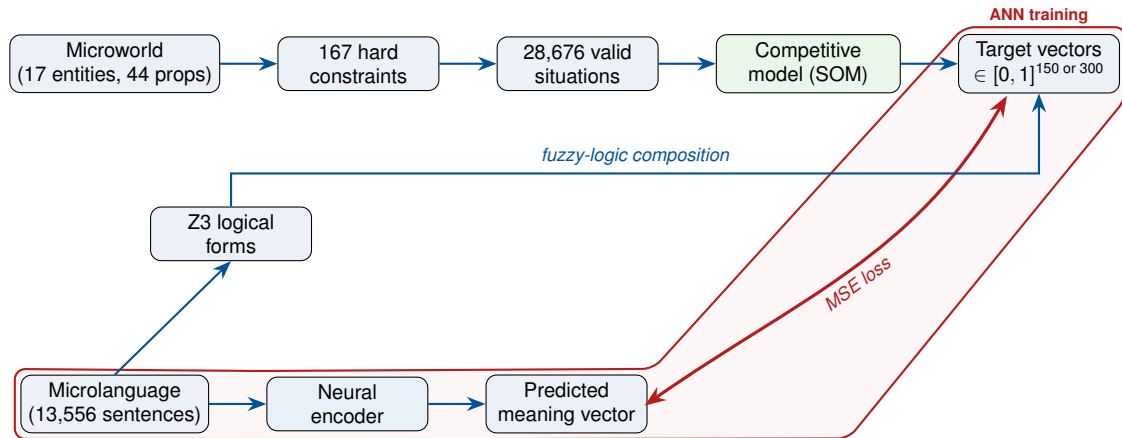
(Frank et al., 2009)

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Framework Overview

Target vector = probabilistic representations of truth conditions the network is trained to output.



Microworld: 17 Entities in 6 Categories

Participants

Persons: charlie, heidi, sophia

Games: chess, hide&seek, soccer

Toys: puzzle, ball, doll

Modifiers

Locations: bathroom, bedroom,
playground, street

Play-manners: well, badly

Win-manners: with-ease, with-difficulty

Specialty System

charlie $\leftrightarrow$ chess

heidi $\leftrightarrow$ hide&seek

sophia $\leftrightarrow$ soccer

Playing your specialty $\rightarrow$
higher win probability

(Frank et al., 2009)

44 Atomic Propositions

A **situation** $\omega \in \{0, 1\}^{44}$ assigns a truth value to each proposition:

Predicate	Example	Count
play(person, game)	play(charlie, chess)	9
play(person, toy)	play(heidi, ball)	9
win(person)	win(sophia)	3
lose(person)	lose(charlie)	3
location(person, place)	location(heidi, bedroom)	12
play(person, manner)	play(sophia, well)	6
win(manner)	win(with-ease)	2
Total		44

167 Hard Constraints → 28,676 Valid Situations

- **Game** (19): at most 1 game per person; player-count requirements
- **Toy** (33): at most 1 toy; soccer needs ball; no toys during chess/hide&seek
- **Location** (72): exactly 1 location per person; co-location for shared games
- **Winning** (31): can't both win & lose; at most 1 winner
- **Manner** (12): at most 1 play manner; requires playing a game

All formalized as Z3 formulas:

$$\neg(c_1 \wedge c_2 \wedge \dots)$$

Drastic reduction

$2^{44} \approx 17.6$ **trillion**
possible binary vectors

↓ 167 hard constraints

28,676 valid situations

Reduction factor ~ 613 million

+ **Soft constraints** shape the probability distribution (playing well → winning more likely)

(Frank et al., 2009; De Moura and Bjørner, 2008; Brasoveanu, 2026)

Truth-Conditional Representations

Pool MCMC sampling (Brasoveanu, 2026) produces representative situations.

A **situation vector** is a continuous, distributed & probabilistic representation of the situations where a proposition is true.

A **competitive neural model** (Self-Organizing Map variant, $n=150$ neurons) learns a weight matrix $\mathbf{W} \in [0, 1]^{150 \times 44}$:

$$\llbracket \varphi_k \rrbracket = \mathbf{W}_{\cdot k} \in [0, 1]^{150} \quad (\text{column } k = \text{meaning of atomic prop } \varphi_k)$$

Complex meanings via differentiable vectorized fuzzy logic:

$$\llbracket \neg \varphi \rrbracket = \mathbf{1} - \llbracket \varphi \rrbracket$$

$$\llbracket \varphi \wedge \psi \rrbracket = \llbracket \varphi \rrbracket \odot \llbracket \psi \rrbracket \quad (\text{element-wise product})$$

$$\llbracket \varphi \vee \psi \rrbracket = \llbracket \varphi \rrbracket + \llbracket \psi \rrbracket - \llbracket \varphi \rrbracket \odot \llbracket \psi \rrbracket$$

Probability estimates: $P(\varphi) = \frac{1}{n} \sum_{i=1}^n \llbracket \varphi \rrbracket_i$
(correlation $r \geq 0.998$ with MCMC sample frequencies)

(Frank et al., 2009; Kohonen, 1995, 2013; Brasoveanu, 2026)

Entity Vectors: Our Extension

Formal semantics treats **entities** (type e) and truth values (type t) as **co-equal primitives**.

Extension: append 17 binary entity-presence indicators to the competitive-model input:

- Entity e is 1 in situation ω if e appears as an argument in any true atomic prop
- Input dimension: $44 \rightarrow 61$
- Competitive model produces $\mathbf{W}^+ \in [0, 1]^{150 \times 61}$

Two training-target conditions:

Condition	Target dim	Content
–ent	150	truth-conditional only
+ent	300	truth-conditional entity-presence

Question: Does making entity information explicit improve compositional generalization?

(Kamp, 1981; Heim, 1982; Brasoveanu, 2026)

Microlanguage

13,556 sentences from a feature-based context-free grammar:

- 40 lexical items
- Active voice: *charlie plays chess*
- Passive voice: *chess is played by charlie*
- Beat / lose: *sophia beats boy, boy loses to sophia*
- Optional modifiers: manner (*plays well*), location (*in bedroom*)

Four synonym pairs for systematicity testing:

charlie ↔ boy	soccer ↔ football
puzzle ↔ jigsaw	bathroom ↔ shower

Existential quantifiers → disjunctions:

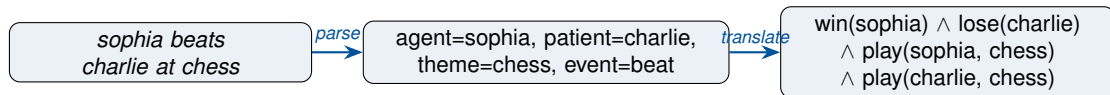
someone → charlie ∨ heidi ∨ sophia

girl → heidi ∨ sophia

(Frank et al., 2009)

Sentence $\rightarrow$ Z3 Logical Form $\rightarrow$ Target Vector

Translation via intermediate event structure (thematic roles):



Target vector: the desired output meaning vector for this sentence.

$$\llbracket \text{sentence} \rrbracket = \llbracket \text{win(sophia)} \rrbracket \odot \llbracket \text{lose(charlie)} \rrbracket \odot \llbracket \text{play(sophia, chess)} \rrbracket \odot \llbracket \text{play(charlie, chess)} \rrbracket$$

After filtering inconsistent sentences:

6,279–6,789 train / 2,330–2,840 test per split

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Seven Neural Architectures

All share the same encoding structure:

word embedding $\rightarrow$ sequence encoder $\rightarrow$ linear projection $\rightarrow$ sigmoid $\rightarrow [0, 1]^n$

Recurrent (Experiment 1)

SRN: Simple Recurrent Network
(Elman, 1990; Frank et al., 2009)

LSTM: Long Short-Term Memory
(Hochreiter and Schmidhuber, 1997; Gers et al., 2000)

GRU: Gated Recurrent Unit
(Cho et al., 2014)

Attention (Exp. 1 & 2)

AbsPE: absolute/sinusoidal positional encoding

RoPE: rotary positional encoding
(Vaswani et al., 2017; Su et al., 2021)

H48/H80: hidden dimension 48/80
Exp. 1 uses H48; Exp. 2 tests higher-capacity H80

140 total models: 7 arch $\times$ $\pm$ entity $\times$ 2 splits $\times$ 5 seeds

Capacity Matching ($\sim 66k$ Parameters)

To isolate **architectural effects** from parameter count:

Architecture	Hidden	Layers	Heads	Params (–ent)
SRN	178	1	—	66,010
GRU	90	1	—	66,210
LSTM	80	1	—	66,950
Attn AbsPE	48	2	4	65,670
Attn RoPE	48	2	4	65,670

Training: AdamW, batch 64, gradient clip 3.0, adaptive LR schedule, 750 epochs

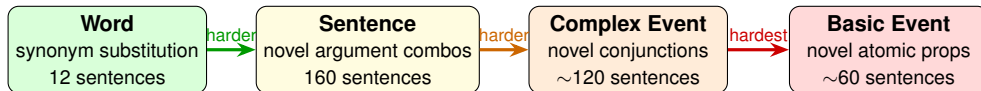
Loss: MSE on full target vector (150 or 300 dims)

Experiment 2: attention with hidden dim **80** (H80, $\sim 170k$ params)

→ tests whether per-position capacity alone explains recurrent advantage

(Loshchilov and Hutter, 2019)

Four Systematicity Tests: Increasing Difficulty



Key distinction: what has the model seen during training?

- **Word:** same truth conditions, different lexical form
- **Sentence:** same truth conditions, different syntactic structure
- **Complex:** individual conjuncts seen, but not their conjunction
- **Basic:** the atomic proposition itself was never a training target

Evaluation Metrics I: Comprehension Score

Comprehension score (Frank et al., 2009): how much the model output $\mathbf{z}$ changes belief in target event $\mathbf{a}$ relative to prior belief.

$$\text{ComprehensionScore}(\mathbf{a} \mid \mathbf{z}) = \begin{cases} \frac{P(\mathbf{a}|\mathbf{z}) - P(\mathbf{a})}{1 - P(\mathbf{a})} & \text{if } P(\mathbf{a} \mid \mathbf{z}) \geq P(\mathbf{a}) \\ \frac{P(\mathbf{a}|\mathbf{z}) - P(\mathbf{a})}{P(\mathbf{a})} & \text{otherwise} \end{cases}$$

- $P(\mathbf{a} \mid \mathbf{z}) = \frac{\frac{1}{150} \sum_{i=1}^{150} (\mathbf{a}_i \cdot \mathbf{z}_i)}{\frac{1}{150} \sum_{i=1}^{150} \mathbf{z}_i}$: probability of target event $\mathbf{a}$ given output vector $\mathbf{z}$.
- $P(\mathbf{a}) = \frac{1}{150} \sum_{i=1}^{150} \mathbf{a}_i$: prior probability of $\mathbf{a}$.
- Comprehension Score is the attained proportion of the maximum possible belief change:
 - ▶ if belief increases, divide by the maximum possible increase $1 - P(\mathbf{a})$;
 - ▶ if belief decreases, divide by the maximum possible decrease $P(\mathbf{a})$.
- Positive Comprehension Score indicates correct support for the described situation; negative Comprehension Score indicates misunderstanding.
- Comprehension Score **always computed on truth-conditional 150-dim part**.

Evaluation Metrics II: Foils and Advantage

Z3 validation: Z3 is a constraint solver; SAT means “possible under the microworld constraints,” UNSAT means “impossible under those constraints.”

Foil events (competing interpretations) via two-step Z3 validation:

- 1 **Validity:** $\text{SAT}(f \wedge \text{constraints})$ —foil event is possible.
- 2 **Competition:** $\text{UNSAT}(f \wedge d \wedge \text{constraints})$ —foil f incompatible with described event d .

Candidate foils vary exactly one event-structure slot at a time: agent, theme, location, or manner.

Advantage score (our primary metric):

$$\text{advantage} = \text{ComprehensionScore}_{\text{described}} - \text{mean}(\text{ComprehensionScore}_{\text{foils}})$$

- High advantage: model supports the described event more than its foils.
- Low or negative advantage: model fails to prefer the intended interpretation.

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Finding 1: Difficulty Has Two Dimensions

Systematicity difficulty $\neq$ single gradient

Dimension 1: Test type

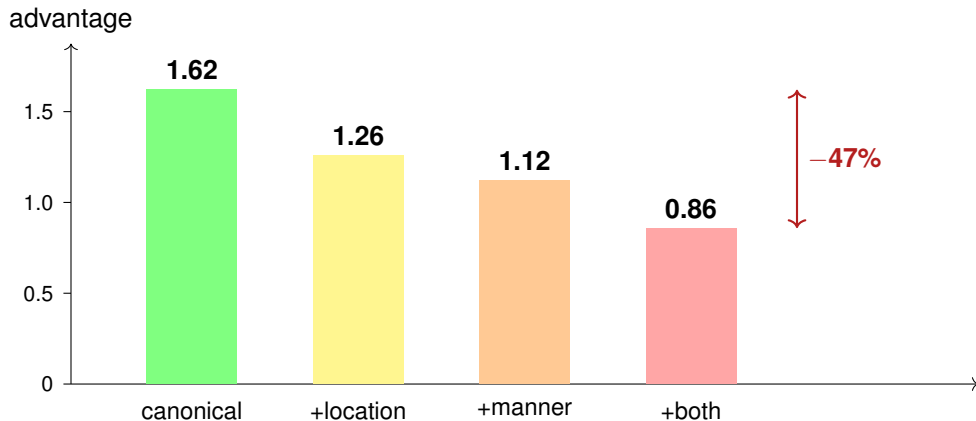
Word > Sentence > Complex Event > Basic Event

Dimension 2: Modifier complexity

Canonical > +Location > +Manner > +Both

Each modifier adds conjuncts to the Z3 formula $\rightarrow$
the model must compose *more* semantic components simultaneously

Modifier Complexity in Action: Sentence Group



sophia beats charlie → *sophia beats charlie in the bedroom*

→ *sophia beats charlie with ease* → *sophia beats charlie with ease in the bedroom*

Systematicity Difficulty: A Two-Dimensional Landscape

	<i>more modifiers to compose</i> →			
	Canonical	+Location	+Manner	+Both
Word	1.94	—	1.58	—
Sentence	1.62	1.26	1.12	0.86
Complex	1.37	—	1.10	—
Basic	0.83	0.65	—	—

Hue = modifier type, as on previous slide; darker shade = higher advantage within column; — = not applicable.

Finding 2: Architecture Effects Across Difficulty Levels

Easy tests (Word): all architectures cluster around ~ 1.7

Hard tests: architecture becomes the dominant factor

- Sentence: $F(6, 139) = 20.72, p < .001$
- Complex: $F(6, 145) = 48.76, p < .001$
- Basic: $F(6, 137) = 7.18, p < .001$

These gaps exist **even at the canonical level:**

- Canonical Sentence: **AbsPE 1.27** vs. **SRN 1.63** and **LSTM 1.69**
- RoPE does well on canonical Sentence (1.72), but not on canonical Complex
- Canonical Complex: **AbsPE 1.16**, **RoPE 1.19** vs. **SRN 1.60**

⇒ The attention deficit is an **intrinsic architectural effect**,
not an artifact of modifier complexity

Finding 3: Gated Recurrent Networks $\gg$ Transformers on Basic Event

Basic Event means (Experiment 1):

	Basic (+ent)	Basic (-ent)
GRU	0.78	0.65
LSTM	0.77	0.67
SRN	0.74	0.67
RoPE	0.69	0.63
AbsPE	0.68	0.59

Pairwise significance statements below use the full 7-architecture analysis, including H80 attention models.

Pairwise contrasts on Basic Event (+ent):

- GRU vs. all 4 attention variants: $p = .001-.034$
- LSTM vs. 3 of 4 attention variants: $p = .002-.034$; vs. RoPE: $p = .053$
- GRU vs. LSTM: $p = 1.00$ (statistically indistinguishable)
- SRN does not significantly outperform attention** in the +ent condition

$\Rightarrow$ **Gating**, not recurrence alone, is the critical factor

But Wait: SRN Excels at Complex Event

	Complex (−ent)	Complex (+ent)
SRN	1.36	1.37
GRU	1.30	1.20
LSTM	1.28	1.24
AbsPE	1.05	1.10
RoPE	1.07	1.05

SRN **significantly outperforms both gated architectures** on Complex Event, and this holds at both Complex modifier levels (canonical and +manner). **Interpretation:** Complex Event tests *sentence-level* composition (recombining familiar conjuncts: activity $\wedge$ location).

- SRN's simpler overwriting dynamics may suit incremental propositional conjunction.
- Gating may not help recombine familiar sentence-level components.

⇒ Gating specifically helps **sub-sentential** composition
(combining words inside an unseen event description; Basic Event)

Finding 4: Entity Vectors Help Most on Basic Event

Basic Event entity benefit (Experiment 1):

Arch	Δ (+ent – –ent)	p
GRU	+0.125	< .001
LSTM	+0.106	< .001
SRN	+0.063	—
AbsPE	+0.096	.002
RoPE	+0.063	—

Overall entity effect on Basic:

$F(1, 189) = 59.17, p < .001$

⇒ Gated architectures benefit most, but the broader pattern is the same:
entity information helps most when composing **novel entity combinations**

Entity effects by test group:

- **Basic:** strong pooled effect ($p < .001$)
- **Word:** pooled positive effect ($p < .001$)
- Sentence: null ($p = .95$)
- Complex: null ($p = .44$)

Entity benefit on Basic is
much larger on test
than on training sentences

(test: +0.094 vs. train: +0.008)

Finding 5: Substantial Generalization Gap

Generalization gap = performance on training sentences minus performance on held-out test sentences.

	Training	Test
Basic Event (all archs)	$\sim 1.01\text{--}1.10$	$\sim 0.57\text{--}0.78$

- Train/test contrast on Basic: $F = 292.45$, $p < .001$
- Entity $\times$ train/test interaction: $F = 67.07$, $p < .001$
→ entity vectors help in both subsets, but **much more on test**
- Complex Event also shows a strong train/test gap
with architecture $\times$ train/test interaction ($F = 44.98$, $p < .001$)

All architectures are ultimately doing fairly poorly on the hardest test.
True compositional generalization **remains challenging**.

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Why Do Gated Recurrent Models Generalize Better?

Three hypotheses tested:

1 Capacity per position?

Recurrent models have larger hidden dims (80–178 vs. 48)

⇒ *Ruled out*: H80 attention ($\sim 170k$ params) doesn't improve

Experiment 2: Attention models w/ hidden dim 80, matching LSTM:

- $\sim 170k$ parameters—**2.5** $\times$ the recurrent models
- does **not** improve over H48 attention (in some cases, slightly *worse*)

2 Sequential inductive bias?

Left-to-right processing, fixed-size hidden state

⇒ *Necessary but insufficient*: SRN shares this but doesn't match GRU/LSTM

3 Gating mechanism ✓

Selective retention and updating of information

⇒ Maintains compositionally relevant features across timesteps

Self-Attention and Information Entanglement

Self-attention computes pairwise interactions between *all* positions:

- Every head mixes “what” (lexical features) with “where/how” (structural roles)
- This may **entangle** object-level and relational information (Webb et al., 2024; Altabaa and Lafferty, 2025)

For compositional semantics, this may be problematic:

- **Lexical semantics** (meanings of words) and **compositional semantics** (how meanings combine) may benefit from *separate processing streams*

One possible interpretation of our results:

- Sequential bottlenecks force compression
- Gates enable selective retention of compositionally relevant features
- Standard self-attention mixes all positions pairwise at every layer

Promising direction: Relational Abstractors, Dual Attention Transformers (Altabaa et al., 2024; Altabaa and Lafferty, 2025)

Implications for Compositionality Research

- **Architectural sophistication $\neq$ compositional ability**
Transformers underperform simpler gated recurrent models on the hardest compositional generalization tests
- Aligns with previous findings: ANNs struggle with compositional generalization (but our setting targets truth conditions, not symbolic transduction)
- **All** architectures are ultimately doing fairly poorly:
best Basic Event advantage = 0.78 (vs. Word $\approx$ 1.7–1.9)
- The right inductive bias matters more than model size for compositional generalization in this paradigm

(Lake and Baroni, 2018; Kim and Linzen, 2020; Keysers et al., 2020)

Implications for the Systematicity Debate

Positive: All architectures exhibit positive systematicity even on Basic Event (0.57–0.78)

- Neural networks *can* learn interpretation functions supporting compositional generalization without classical symbols

But: Systematicity is **graded**, not all-or-nothing

- Near-ceiling on synonym substitution in canonical sentences
- Very modest on novel atomic propositions with modifiers

The neural systematicity landscape:

- Complex interaction of architectural inductive biases, representation richness, and compositional complexity

Code & data: https://github.com/abrsvn/neural_model_theoretic_interpretation_ACL_2026

References I

- Altabaa, A. and Lafferty, J. (2025). Disentangling and integrating relational and sensory information in transformer architectures. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 1271–1297. PMLR.
- Altabaa, A., Webb, T., Cohen, J. D., and Lafferty, J. (2024). Abstractors and relational cross-attention: An inductive bias for explicit relational reasoning in transformers. In *ICLR*.
- Brasoveanu, A. (2026). Distributed situation models: Probabilistic representations of truth-conditions. In *Proceedings of the Annual Meeting of the Cognitive Science Society*, volume 48.
- Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., and Bengio, Y. (2014). Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1724–1734.
- De Moura, L. and Bjørner, N. (2008). *Z3: An efficient SMT solver*. Springer.
- Elman, J. L. (1990). Finding structure in time. *Cognitive Science*, 14(2):179–211.
- Fodor, J. A. and Pylyshyn, Z. W. (1988). Connectionism and cognitive architecture: A critical analysis. *Cognition*, 28(1-2):3–71.
- Frank, S. L., Haselager, W. F., and Van Rooij, I. (2009). Connectionist semantic systematicity. *Cognition*, 110(3):358–379.
- Gers, F. A., Schmidhuber, J., and Cummins, F. (2000). Learning to forget: Continual prediction with LSTM. *Neural Computation*, 12(10):2451–2471.
- Heim, I. (1982). *The Semantics of Definite and Indefinite Noun Phrases*. PhD thesis, University of Massachusetts, Amherst.
- Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8):1735–1780.
- Kamp, H. (1981). A theory of truth and semantic representation. In Groenendijk, J., Janssen, T., and Stokhof, M., editors, *Formal Methods in the Study of Language*, pages 277–322, Amsterdam. Mathematical Centre.
- Keysers, D., Schärli, N., Scales, N., Buisman, H., Furrer, D., Kashubin, S., Momchev, N., Sinopalnikov, D., Stafiniak, L., Tihon, T., Tsarkov, D., Wang, X., van Zee, M., and Bousquet, O. (2020). Measuring compositional generalization: A comprehensive method on realistic data. In *International Conference on Learning Representations*.
- Kim, N. and Linzen, T. (2020). COGS: A compositional generalization challenge based on semantic interpretation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9087–9105. Association for Computational Linguistics.
- Kohonen, T. (1995). *Self-Organizing Maps*, volume 30 of *Springer Series in Information Sciences*. Springer, Berlin, Heidelberg.

References II

- Kohonen, T. (2013). Essentials of the self-organizing map. *Neural Networks*, 37:52–65.
- Lake, B. and Baroni, M. (2018). Generalization without systematicity: On the compositional skills of sequence-to-sequence recurrent networks. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 2873–2882. PMLR.
- Loshchilov, I. and Hutter, F. (2019). Decoupled weight decay regularization. In *International Conference on Learning Representations*.
- Su, J., Lu, Y., Pan, S., Wen, B., and Liu, Y. (2021). Roformer: Enhanced transformer with rotary position embedding. *arXiv preprint arXiv:2104.09864*.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, pages 5998–6008.
- Webb, T. W., Frankland, S. M., Altabaa, A., et al. (2024). The relational bottleneck as an inductive bias for efficient abstraction. *Trends in Cognitive Sciences*.